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Centimeter-sized 2D α-MoO_3 single crystal: growth, Raman anisotropy, and optoelectronic properties

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Abstract

Centimeter-sized 2D layered α-MoO_3 single crystal has been grown by a facile physical vapor deposition method. The lateral size of α-MoO_3 single crystals can be tunable from micrometer to centimeter by controlling the inner diameter of the quartz tube reaction chamber. The as-grown α-MoO_3 exhibits strong in-plane Raman anisotropy, where A_g Raman modes give the direction of crystallographic orientation, which can be used to rapidly and precisely identify the crystallinity of large-area α-MoO_3 through selected-area angel-resolved polarized Raman spectroscopy. Moreover, the photodetectors based on annealed α-MoO_3 single crystals exhibit excellent optoelectronic performance with a photoresponsivity of 65.6 A/W and a fast photoresponse time of 2 s.

Keywords: two-dimensional semiconductor, physical vapor deposition, molybdenum trioxide, Raman anisotropy, photoresponse

1. Introduction

Two-dimensional (2D) materials with layered structure, unusual electronic and optical properties have attracted plenty of attention, becoming a huge hit in this decade since the emergence of graphene [1-4]. Significant research efforts have been invested to the family of 2D materials, including 2D transitional metal dichalcogenides (TMDs, such as MoS_2 and $MoSe_2$) [5], 2D transition metal carbides, carbonitrides and nitrides (MXenes, such as Ti_2C and Ti_3C_2) [6], 2D monoelemental materials (such as graphene and black phosphorus) [7], and the latest 2D transition metal oxides (TMOs) [8, 9]. As a member of the 2D TMOs, orthorhombic molybdenum trioxide (α-MoO_3) is an amazing layered semiconductor with wide bandgap. α-MoO_3 single crystals can be mechanically exfoliated into few-layered nanosheets, which can be widely applied in field-effect transistor [10, 11], synaptic transistor [12], photodetector [13-15], biosensor [16, 17], gas sensor [18], and electrocatalysis [19, 20]. It is well-known that mechanical exfoliation, utilizing easy cleavage of 2D crystal at its weak point along the van der Waals' gap [21], is a facile and desirable method to obtain pristine, atomically thin 2D nanosheets with purity, high crystallinity and low defect density. However, it is very hard to synthesize the single crystal starting material (e.g. HOPG for graphene) with large size and high crystalline quality for mechanical exfoliation. Generally, the starting materials are synthesized

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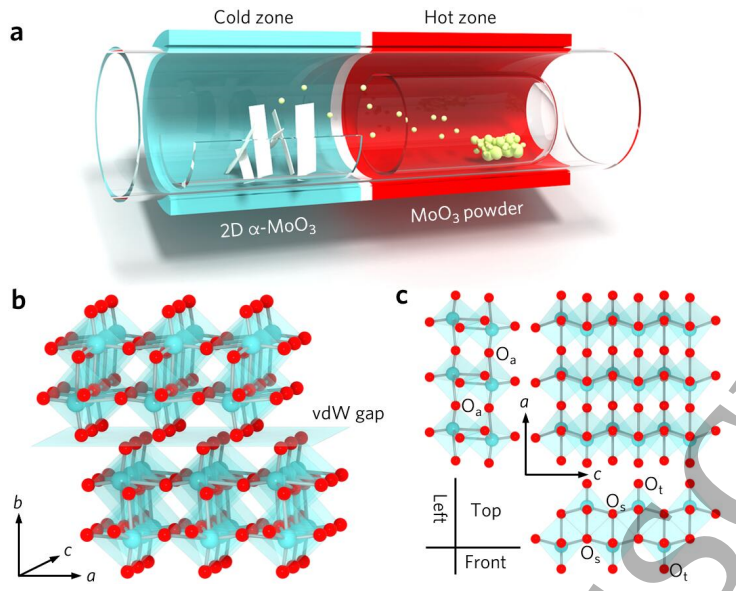

Figure 1. Synthesis and crystal structure of α-MoO_3 single crystal. (a) Schematic illustration of synthesis of α-MoO_3 in a two-temperature-zone PVD system. (b) Crystal structure of α-MoO_3. (c) Three views of monolayer α-MoO_3.

by the chemical vapor transport (CVT) method, which is time-consuming and expensive [22, 23]. Apparently, it is significant to synthesize the starting material with high crystalline quality and large area to mechanically exfoliate 2D flakes for easy characterization and device fabrication, and good device performance. To the best of our knowledge, the direct growth of α-MoO_3 single crystal is far from satisfactory. So far, the largest lateral size of α-MoO_3 single crystal ever reported is only a few tens of micrometers [14, 15, 24]. Therefore, it is urgent to develop a facile method to synthesize α-MoO_3 single crystals with high quality and large area.

Herein, we present a facile physical vapor deposition (PVD) method with the use of a tube furnace set-up to synthesize centimeter-sized α-MoO_3 single crystals with high crystalline quality at atmospheric pressure. The lateral size of α-MoO_3 single crystals can be significantly enlarged from micrometer to centimeter by decreasing the inner diameter of the quartz tube reaction chamber from 4.2 cm down to 2.1 cm. Furthermore, as-grown α-MoO_3 exhibits strong in-plane Raman anisotropy, which can be used to rapidly identify the crystallinity of α-MoO_3 crystal through selected-area angel-resolved polarized Raman spectroscopy (SAPRS). Moreover, the photodetectors based on as-synthesized and annealed α-MoO_3 single crystals exhibit excellent optoelectronic performance.

2. Results and discussion

High-quality α-MoO_3 single crystal were synthesized via PVD method at atmospheric pressure as illustrated in **figure 1**a. Briefly, a quartz test tube loaded with MoO_3 powder, used as a directional evaporator source, was placed at the hot zone of furnace, while a quartz boat used as a 2D α-MoO_3 holder was placed at the cold zone. The structural framework of α-MoO_3 is shown in figure 1b. Monolayer α-MoO_3 consists of a bilayer network of [MoO_6] octahedra, which share corners along the *a*-axis ([100]) and share edges along the *c*-axis ([001]). The layers stack up through weak van der Waals' force along the *b*-axis ([010]), which offers the possibility to be exfoliated into atomically thin sheets. The Mo to Mo atom distance between two layers is 1.4 nm [10]. Figure 1c clearly shows the three types of oxygen atoms [25]. The terminal oxygen (O_t), in both top and bottom surfaces of monolayer, forms Mo-O_t bond in *b* direction. The asymmetric oxygen (O_a) forms bonds with two Mo atoms in the *a*-axis. And, the symmetric oxygen (O_s), forms two short bonds with two Mo atoms in the *c*-axis and a longer bond with a Mo atom in the *b*-axis.

Further investigation of the effects of growth conditions on sheet size of α-MoO_3 single crystals was performed. Given that the as-grown α-MoO_3 has a quasi-rectangular shape, length and width are used to describe the sheet size. **Figure 2**a-b show the remarkable increase of sheet size when the tube inner diameter decreases. Whereas the variations of evaporating temperature and source mass do not have as much leverage. Lower or higher temperature leads to a fall in quantity of large sheets (figure 2d, c, e), while larger and smaller mass result in fragmentation and shortening of sheets, respectively (figure 2f, c, g). The statistical analysis of sheet size evolution is depicted in figure 2h. When the tube inner diameter decreases from 4.2 to 3 cm, the average sheet size enlarges 3 orders of magnitude. Continuing decreasing the tube inner diameter to 2.1 cm, the average width increases from 0.02 to 0.15 cm, but the average length grows a little from 0.68 to 0.72 cm. Each decrease of inner diameter halves the cross-sectional area, i.e., doubles MoO_3 vapor concentration, resulting in enlargement of MoO_3 sheet, while

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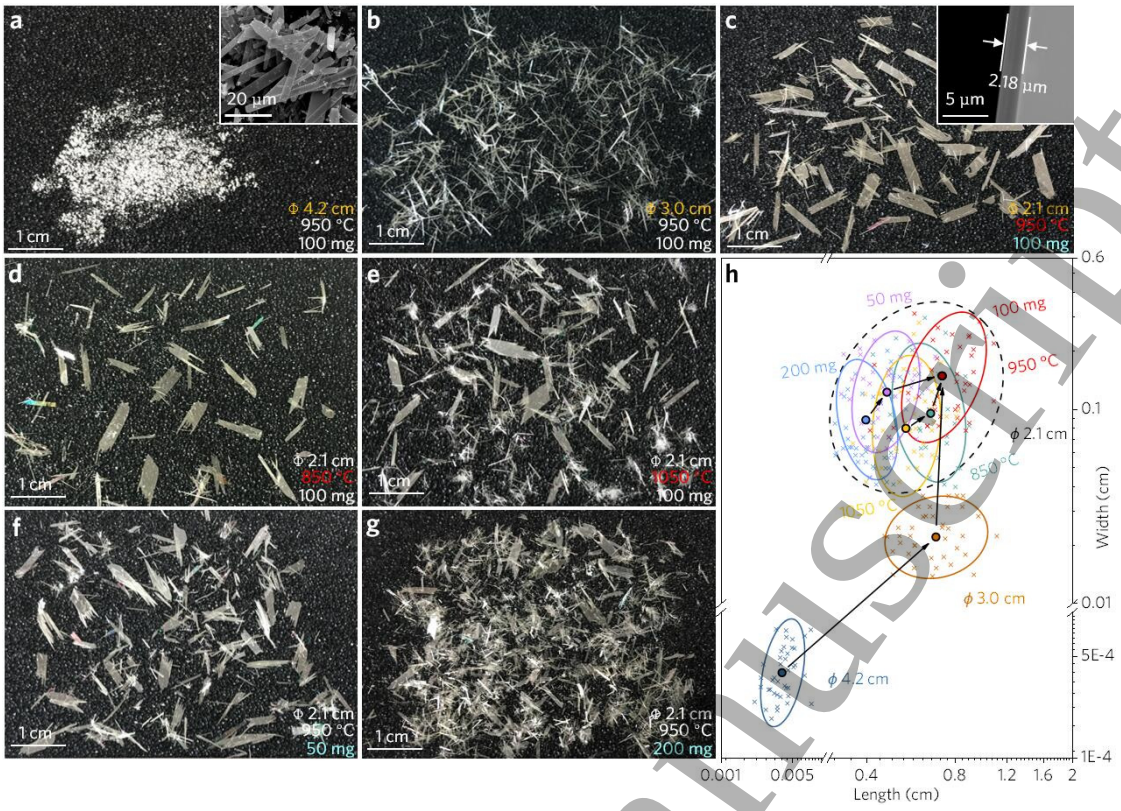

Figure 2. Sheet size evolution of α-MoO_3. (a, b, c) Photographs of α-MoO_3 synthesized with tube inner diameter of 4.2 cm, 3 cm and 2.1 cm. insets: corresponding SEM images (c, d, e) Photographs of α-MoO_3 synthesized with evaporating temperature of 950°C, 850 °C and 1050 °C. (c, f, g) Photographs of α-MoO_3 synthesized with source mass of 100 mg, 50 mg and 200 mg. (h) Size statistical distribution of different parameters.

the length of sheet is limited to the inner radius of tube. The raise of temperature and source mass will enrich MoO_3 vapor concentration while simultaneously accelerate MoO_3 vapor velocity, causing that the laminar flow may be disturbed and may become turbulent which is unfavorable for crystal growth. In conclusion, tuning the inner diameter of tube is an effective way to control the morphology and size of single crystal, which might potentially be applied to grow other 2D materials.

The typical optical image of as-grown α-MoO_3 single crystal collected from figure 2c on SiO_2/Si substrate is exhibited in **figure 3**a. All four centimeter-scale sheets represent an almost rectangular shape, implying it is a preferred-oriented growth. According to the periodic bond chain (PBC) theory the crystal habit is governed by a set of uninterrupted chains of strong bonds formed in the crystal lattice [26]. The PBC in the *b*-axis is consist of strong O_t-Mo-O_s covalent bond and weak van der Waals, thus the O_t-Mo-O_s bond has no influence on the shape of the crystal and (010) face will become the largest face. The PBCs in the *a*-axis and *c*-axis are formed by -O_a-Mo- bonds and -O_s-Mo- bonds, respectively. Because the latter is stronger we assume the crystal will be elongated into the direction [001], that is, the lengthwise direction should be *c*-axis of the α-MoO_3. It is confirmed through High-resolution transmission electron microscopy (HRTEM) shown in figure 3b. Two sets of parallel lines indicate lattice constants of 0.398 and 0.375 nm, respectively, which are consistent with the interplanar distances of (100) and (001) lattice planes of α-MoO_3 [15]. The corresponding fast Fourier transform (FFT) in inset shows a rectangular position of diffraction spots, indicating the orthorhombic lattice. Combining the TEM image at low magnification, it reveals that the lengthwise direction of α-MoO_3 rectangular sheet is along the *c*-axis, while the widthwise direction is along the *a*-axis. The stoichiometry of as-grown α-MoO_3 single crystal was determined by X-ray photoelectron spectroscopy (XPS). As shown in figure 3c, for Mo 3d spectrum, two peaks located at 232.49 and 235.63 eV are ascribed to Mo^{6+} $3d_{5/2}$ and $3d_{3/2}$, respectively [20]. For O 1s spectrum, the main peak at 530.5 eV is assigned to the crystal lattice oxygen, while the shoulder peak at 532.28 eV can be assigned to adsorbed water [17]. The atomic ratio of Mo: O is 2.97, indicating the as-grown α-MoO_3 is stoichiometric.

X-ray diffraction (XRD) was utilized to characterize the crystallinity of as-grown α-MoO_3. As shown in figure 3d, only (0k0) diffraction patterns, i.e. 12.7° (020), 25.6° (040), 38.9° (060), 52.7° (080) and 67.6° (0100) are observed in XRD ω-2θ scan. The observation of only (0k0) peaks indicates that the b-axis of α-MoO_3 is perpendicular to the SiO_2/Si substrate, which means the layers of α-MoO_3 are parallel to the substrate. Meanwhile, XRD rocking curve measurements were also

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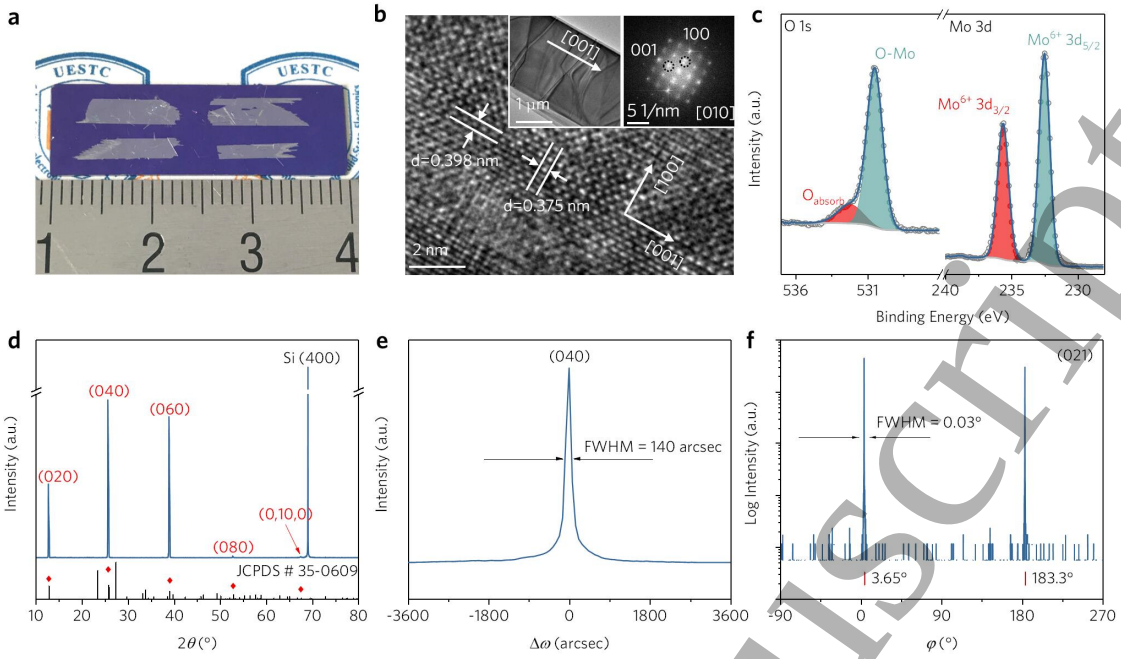

Figure 3. The crystal orientation and composition of the as-grown α-MoO_3 single crystals. (a) Photograph of as-grown α-MoO_3 single crystals on SiO_2/Si substrate. (b) HRTEM image of the α-MoO_3 sheet. Insets: corresponding TEM and FFT images. (c) High-resolution XPS spectra of O 1s and Mo 3d of as-grown α-MoO_3. XRD (d) 2θ-ω scans, (e) rocking curve of the (040) and (f) azimuthal (φ) scan of the (021) of α-MoO_3 crystal on SiO_2/Si substrate.

performed to investigate the deviation of the α-MoO_3 crystallite axes form the normal direction of the surface. The rocking curve of the (040) reflection shown in figure 3e indicates a full width at half maximum (FWHM) as narrow as 140 arcsec. Such a sharpness of the rocking curve is astonishing in layered materials, even better than that grown by CVT method [23]. Furthermore, to investigate the in-plane epitaxial relationship, XRD azimuthal scan was obtained for the (021) reflection of α-MoO_3 crystal. As shown in figure 3f, two strong and sharp peaks are observed at 180° intervals, clearly indicating two-fold mirror symmetry. XRD results demonstrate the as-grown α-MoO_3 is single-crystal and highly crystalline.

Recently, Raman anisotropy has been (re)discovered not only in 2D materials with in-plane low symmetry, such as black phosphorus [27], Re(S,Se)$_2$ [28], SnS [29] and $MoTe_2$ [30], but also in isotropic 2D materials, such as graphene [31, 32], (Mo, W)(S, Se)$_2$ [33], which can be used to reveal the crystallographic orientation of measured area of specimen. Thus, a systematic investigation on Raman characterization of as-grown α-MoO_3 single crystals was performed. α-MoO_3 belongs to the D_{2h}^{16} space group, and there are 48 lattice vibrational modes at the Γ point [34], where A_g, B_{1g}, B_{2g}, and B_{3g} are Raman-active modes. Coordinate system and polarization configuration are shown in **figure 4**a, black arrows (*xyz*) indicate the laboratory coordinate; green arrows (*abc*) stand for the crystal coordinate; red two-way arrow represents the incident laser polarization reaching at sample; blue two-way arrow indicates the analyzer direction. This configuration is widely used in other anisotropic layered materials [27-29]. Although as-grown α-MoO_3 has a rectangle-like shape and the lengthwise direction is the *c*-axis, it is necessary to consider the generalized condition where crystalline orientation is unknown. We set an initial angle α (~14°, measured from optical image) between sample crystal coordinate and laboratory coordinate. Thus, when the sample was clockwise rotated in the *x*-*y* plane by an angle θ, there is an angle $\beta = \theta - \alpha$ between laboratory coordinates and crystal coordinates.

The intensity of a given Raman active mode can be calculated by $I \propto \left|e_i \cdot R \cdot e_s\right|^2$, where R is a Raman tensor, four Raman active phono modes can be expressed as [35]:

$$R(A_g) = \begin{pmatrix} A & 0 & 0 \\ 0 & B & 0 \\ 0 & 0 & C \end{pmatrix},\ R(B_{1g}) = \begin{pmatrix} 0 & D & 0 \\ D & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$R(B_{2g}) = \begin{pmatrix} 0 & 0 & E \\ 0 & 0 & 0 \\ E & 0 & 0 \end{pmatrix},\ R(B_{3g}) = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & F \\ 0 & F & 0 \end{pmatrix}$$

e_i and e_s are the unit polarization vectors of the incident laser and scattered Raman signal, respectively. The incident laser propagates along *y*-axis with polarization vector along *x*-axis: $e_i = \begin{pmatrix} 1 & 0 & 0 \end{pmatrix}$. The Raman signal is scattered along *y* axis with polarization vector fixed by the analyzer: $e_s = \begin{pmatrix} 1 & 0 & 0 \end{pmatrix}^T$. Because the Raman tensor (R) of a crystal is defined according to its crystal coordinates, it must be converted into laboratory coordinates (R') by $R' = tRt^T$, where

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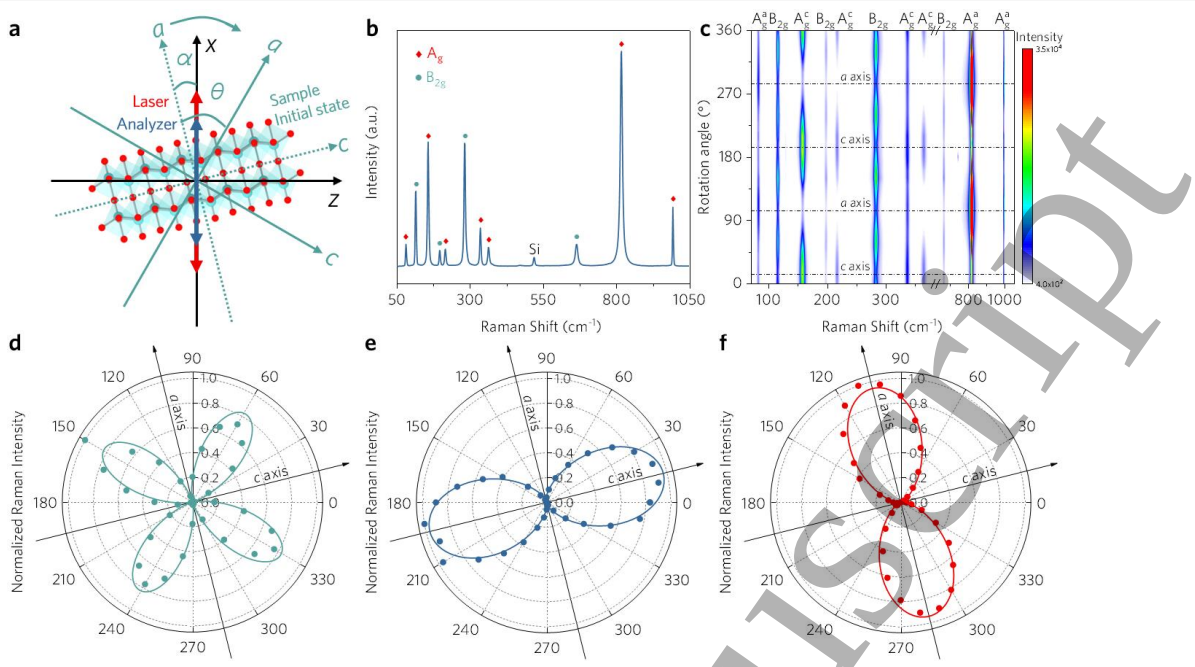

Figure 4. Angle-resolved polarized Raman analysis of α-MoO_3 crystal. (a) Schematic diagrams of the configuration in angle-resolved polarized Raman experiments. (b) Typical Raman spectrum of α-MoO_3 sample. (c) False-color plot of polarized Raman intensity of α-MoO_3 sample. (d, e, f) corresponding polar plots (dots) and fittings (lines) of normalized Raman intensities of three typical peaks (283 cm^{-1}, 158 cm^{-1} and 818 cm^{-1}) as function of sample rotation angle θ.

t is defined as a transform matrix linking the two coordinates [36]:

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = t \begin{pmatrix} a \\ b \\ c \end{pmatrix},\ t = \begin{pmatrix} \cos\beta & 0 & -\sin\beta \\ 0 & 1 & 0 \\ \sin\beta & 0 & \cos\beta \end{pmatrix}$$

Thus, Raman intensity is evolved into

$$I \propto \left| e_i \cdot t \cdot R \cdot t^{\mathrm{T}} \cdot e_s \right|^2$$

And intensities of four Raman active modes can be written as:

$$I\left(\mathrm{A_g}\right) \propto \left(A\cos^2\beta + C\sin^2\beta\right)^2 \quad (1)$$

$$I\left(\mathrm{B_{2g}}\right) \propto E^2 \sin^2 2\beta \quad (2)$$

$$I\left(\mathrm{B_{1g}}\right) = I\left(\mathrm{B_{3g}}\right) = 0$$

which indicate that only A_g and B_{2g} modes can be detected, while the intensity variation period of A_g mode is 180°, that of B_{2g} is 90°.

The typical Raman spectrum of as-grown α-MoO_3 is shown in figure 4b, only two modes of Raman are observed. Peaks at 82, 158, 216, 336, 364, 818 and 994 cm^{-1} are assigned as the A_{1g} mode and peaks at 115, 197, 283 and 665 cm^{-1} are as the B_{2g} mode. By rotating the sample from 0° to 360°, polarized Raman intensity was mapped in figure 4c, illustrating strong angle-dependence and Raman-mode dependence. Raman peaks belong to B_{2g} mode show four maximum intensities along directions between *a*- and *c*-axes. It is of interest to notice two kinds of A_g modes: peaks at 158, 216, 336 and 364 cm^{-1} show two maximum intensities along *c*-axis (represented by A_g^c); peaks at 82, 818 and 994 cm^{-1} show two maximum intensities along *a*-axis (represented by A_g^a). It is attributed to the different values of *A* and *C* in the Raman tensor [37], resulting the local maximum A^2 or C^2 with the difference of 90°. To further investigate the Raman anisotropy of as-grown α-MoO_3 crystal, we focused on three typical Raman peaks located at 283, 158 and 818 cm^{-1}, which respectively belong to B_{2g}, A_g^c and A_g^a Raman modes with more evident intensity variation observed in figure 4c. The corresponding polar plots and fittings of normalized Raman intensities of these Raman peaks as function of sample rotation angle θ are shown in figure 4d-f. It is clearly that different Raman modes have different periodic variation features, while the experimental data are well fitted by equatin 1 and equation 2. The B_{2g} mode has a variation period of 90°, which reaches the local maximum at θ= 59°, 149°,239° and 329°. Considering the initial angle α is about 14°, the B_{2g} mode shows the local maximum at a 45° on either side of *a*- or *c*-axes, which cannot be used to indicate the crystalline orientation, unfortunately. However, both A_g^c and A_g^a modes have a 180° variation period, reaching the local maximum at *c*- axis and *a*-axis while being forbidden at *a*-axis and *c*-axis, respectively. That means both A_g^c and A_g^a modes can be used to ascertain the specific direction of the crystalline orientation. Essentially, this is because the Raman peak at 158 cm^{-1} is translation vibration of the rigid MoO_6 octahedra chains along the *c*-axis, and the Raman peak at 818 cm^{-1} is asymmetric stretching vibration of O_a-Mo-O_a along the *a*-axis [34]. The calculated and

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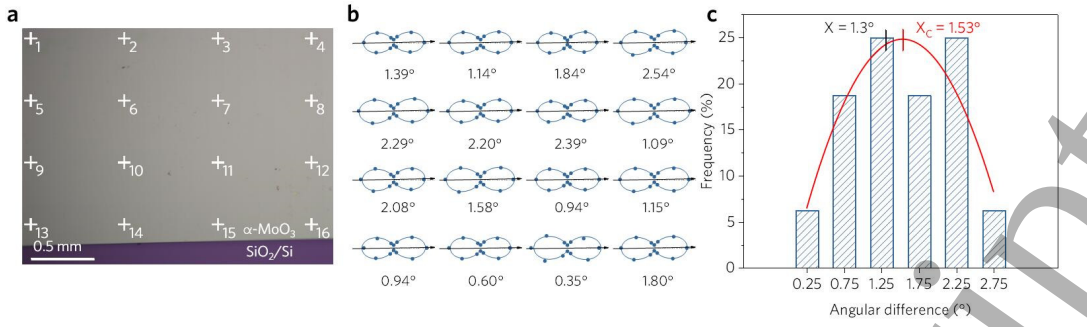

Figure 5. Selected-area angel-resolved polarized Raman spectroscopy (SAPRS) of the as-grown α-MoO_3 single crystals. (a) optical image of α-MoO_3 on SiO_2/Si substrate. 16 test points are labelled with cross symbols and numbers. (b) SAPRS plots (158 cm^{-1}) taken form the areas labelled with numbers 1-16 in (a). The angles of *c*-axes with respect to horizontal direction were labelled below plots. (c) Angle-difference comparison of experimental data and theoretical value.

experimental results further demonstrate the optical anisotropic nature of as-grown α-MoO_3.

The crystal orientation and single crystallinity of the as-grown α-MoO_3 single crystals on SiO_2/Si substrate were further investigated and confirmed utilizing the SAPRS. A series of SAPRS measurements (158 cm^{-1}) at 16 different points (diameter of ~1 μm, **figure 5**a) were performed, as presented in figure 5b. Each SAPRS plot only shows one figure-eight pattern, indicating one direction of *c*-axes. If the SAPRS pattern is a circle, it infers that tested area is amorphous or non-oriented nanocrystalline; if the SAPRS pattern is figure-eight and different from adjacent SAPRS patterns, it infers that tested area is polycrystal; if all the SAPRS patterns are figure-eight and almost all the same, it infers that the tested sample is single crystal. In figure 5b, the angles of *c*-axes with respect to horizontal direction were labelled below plots. Given that the precision of instrument is 2°, all the SAPRS plots show the perfect coincidence with deviation smaller than ±1°. Figure 5c shows the average angle of SAPRS data is 1.53°, which is in good consistent with the theoretical *c*-axis (lengthwise direction of rectangle, 1.3° from horizontal direction). Over all, SAPRS plots precisely confirm single crystallinity of the as-grown α-MoO_3, which is similar to the feature of SAED patterns [38, 39], and SARPS has its unique features of transfer-free, non-vacuum, non-destructive, rapid and suitable for large-area characterization.

As a typical application example, here the photodetectors based on α-MoO_3 single crystals were fabricated. **Figure 6**a shows the *I-V* curves of the device measured in the dark and under 405-nm laser irradiation. When the device was

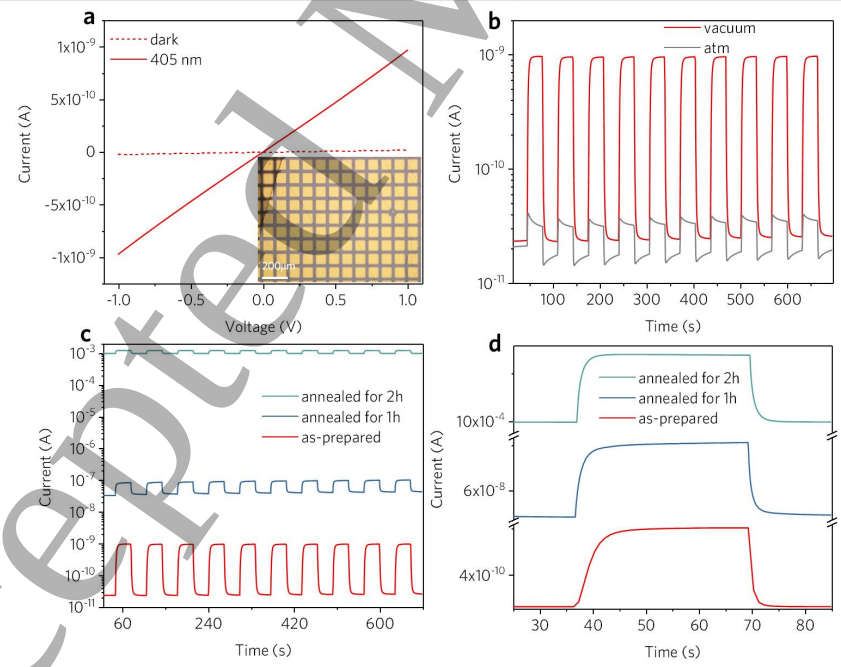

Figure 6. Optoelectronic properties of α-MoO_3 single crystal. (a) *I-V* curve of as-grown-sample-based photodetector measured in the dark and under 405-nm laser irradiation. Inset: optical image of photodetectors. (b) Time-resolved photoresponse of as-grown-sample-based photodetector measured in vacuum and atmosphere. (c) Time-resolved photoresponse of the device with different annealing time. (d) Enlarged portion of one single photoresponse cycle.

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illuminated, a drastic increase of current from 2.34×10^{-11} A up to 9.65×10^{-10} A at voltage of 1 V exhibits high signal-to-noise ratio of ca. 10^2, indicating the good optoelectronic properties of α-MoO_3 single crystal. Besides, the good and symmetric linearity implies that ohmic nature of the Au-MoO_3 contact. Figure 6b shows the time-resolved photoresponse of α-MoO_3 photodetector measured in vacuum and atmosphere. It is clearly that the device measured in atmosphere display a poor performance and a phenomenon of transient photocurrent response [40]. The poor performance is caused by that the adsorbed oxygen molecules capture free electrons of α-MoO_3 to form adsorbed oxygen anions [41]. The instantaneous current when the light is switched on is caused by that the flus of photogenerated holes into the surface combine with adsorbed oxygen anions to free the trapped carriers, and vice versa [40, 41]. Figure 6c shows the long-term stability and repeatability of photodetectors based on α-MoO_3 single crystals with different vacuum annealing time at 200°C. With the increase of annealing time, the signal-to-noise ratio declines and both the dark current (I_d) and illumination current (I_i) increase due to the enhancement of conductivity [10, 11, 13, 15]. The photoresponsivity is a measure of the electrical response to light and is calculated by $R= (I_i- I_d)/(P\times S)$, where P is the power density of light source, S is the effective illuminated area of device[42]. The photodetectivity is related to the sensitivity that a detector can distinguish from the background noise and is calculated by $D^*= RS^{1/2}/(2q\ I_d)^{1/2}$ [42], where q is the unit of charge. The photodetector based on as-grown α-MoO_3 single crystals have a responsivity of 2.62×10^{-4} A/W and a detectivity of 1.15×10^7 Jones. After annealing for 1 h (I_d=3.28×10^{-8} A, I_i=8.35×10^{-8} A), the responsivity increases 2 orders of magnitude to 1.41×10^{-2} A/W, but the detectivity increases slightly to 1.65×10^7 Jones because the reduction of signal-to-noise ratio offsets the increase of photocurrent. After annealing for 2 h (I_d=9.84×10^{-4} A, I_i=1.22×10^{-3} A), the responsivity and detectivity increase to 65.6 A/W and 4.43×10^8 Jones, respectively, which is better than Annealed α-MoO_3 nanowire [13] and nanosheet [14, 15]. The enlarged portion of one single photoresponse cycle is given in figure 6d. The photoresponse time of photodetector based on as-grown α-MoO_3 is 5 s, which is 1 order of magnitude faster than the α-MoO_3 crystals grown by other strategies [13-15], revealing the high crystallinity, low defects and low trap states of as-grown α-MoO_3 single crystal. Moreover, with increase of annealing time, the photoresponse speed is further improved to 2 s, which is due to the increase of photogenerated carrier mobility [10, 11].

3. Conclusions

Centimeter-sized α-MoO_3 single crystals have been synthesized using a facile PVD method at atmospheric pressure. It is effective to control the morphology and size of single crystal by varying the inner diameter of the quartz tube reaction chamber, source mass and evaporation temperature. The lateral size of single crystals can be significantly enlarged from micrometer to centimeter, which might potentially be applied to grow other 2D materials. The XPS and XRD data show that the as-grown α-MoO_3 is single crystal with high crystallinity and chemical stoichiometry. The α-MoO_3 has strong in-plane Raman anisotropy, where A_g^c and A_g^a Raman modes can be used to identify the c-axis and a-axis crystallographic orientation, respectively. Furthermore, SAPRS patterns can be used as a transfer-free, non-vacuum, non-destructive, approach to rapidly and precisely identify the crystallinity of large-area α-MoO_3 crystal. The perfect coincidence (deviation smaller than ±1°) of all the SAPRS patterns confirms the single crystallinity of the as-grown α-MoO_3. It is promising that SAPRS patterns can be applied to investigate not only the crystallinity, but also the epitaxial relationship, the crystalline orientation of the heterostructure, and the stacking order of 2D materials with a wide range of size. The photodetector based on as-grown α-MoO_3 single crystal exhibit fast photoresponse time of 5 s due to the high crystallinity, low defects and low trap states. Furthermore, with the increase of vacuum annealing time, the responsivity, detectivity, and response time can be significantly enhanced to 65.6 A/W, 4.43×10^8 Jones, and 2 s, respectively, suggesting promising application in optoelectronics.

4. Experimental

4.1 synthesis of α-MoO_3 single crystal. To prevent sub-stoichiometry MoO_{3-x} caused by vacuum, the PVD was performed at atmospheric pressure with oxygen. A quartz test tube (8 mm in inner diameter, 3 cm in length) loaded with MoO_3 powder (99.95% metals basis, Alfa Aesar), used as a directional evaporator source, was placed at the hot zone of furnace, while a long quartz boat used as a 2D α-MoO_3 holder was placed at the cold zone (12 cm long, wrapped by heat insulation cotton). The distance between source and holder was 15 cm. The hot zone was heated to target temperature at the rate of 25°C min^{-1} and kept for 30 min. After growth, the heating was stopped, and the furnace was opened immediately to cool down.

4.2 Characterizations. The crystallinity analysis was examined by transmission electron microscope (TEM; FEI Tecnai F20) and X-ray diffraction (XRD; Jordan Valley D1 Evolution). The Stoichiometry was performed by X-ray photoelectron spectroscopy (XPS; Kratos XSAM800). Raman spectra were collected on a Raman spectrometer (Renishaw inVia Reflex) with 532 nm laser. The laser power on the sample was 1.5 mW while the spot size was ~1 μm. The silicon peak at 520.5 cm^{-1} was used as a reference for wavenumber calibration. For selected-area angel-resolved polarized Raman spectroscopy (SAPRS), a rotation stage (SELN RSP40-L) was equipped on the motion platform of

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Raman microscopy. Keep the measured spot of sample exactly at the rotation axis. If you have a half-wave plate, it will be more easier to perform SAPRS measurement [36].

4.3 Device fabrication. To fabricate the α-MoO_3 photodetectors, gold square electrodes with 80 μm on a side and spacing of 20 μm were patterned by magnetron sputtering using TEM Cu grid as metal hard mask. The effective illuminated area of device is 1.44×10^{-8} m^2. The electrical characteristics of fabricated devices were carried out on a Lakeshore probe station using a Keithley 2636B source meter. The light source was a commercial laser pointer with wavelength of 405 nm, power of 5 mW and power density of 250 W/m^2.

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